OMAR TAWFIK

+1 (240) 494-6049 | omaressam390@gmail.com | https://www.linkedin.com/in/omar-tawfik123/ | https://github.com/tej-jeenu

EDUCATION

University of Maryland - College Park **August 2024 - May 2026**
Master's, Data Science *GPA: 4*

University of Maryland - College Park **August 2019 - May 2023**
Bachelor's, Business Analytics *GPA: 3.9*

PROFESSIONAL EXPERIENCE

Office of Labor Relations & Collective Bargaining (OLRCB) **District of Columbia Government**
Labor Economist / Data Scientist *January 2026 – Present*

- Led labor economics and compensation analyses across 5+ active bargaining and settlement matters, using Python, SQL, and advanced statistical modeling to evaluate proposals affecting 1,200–1,800 unionized employees and $12–$20M in projected fiscal exposure.
- Built multi-year forecasting and scenario models leveraging time-series analysis, regression modeling, and stochastic sensitivity analysis to assess wage growth, benefit costs, and settlement risk across 3–5 negotiation strategies.
- Developed defensible back-pay and compensation models using structured payroll, leave, and demographic data, applying feature engineering and reproducible modeling pipelines to evaluate $4M in potential settlement liability.
- Established a centralized analytics and compliance framework using SQL-based data models and audit-ready workflows, tracking 20+ compensation-related obligations arising from CBAs, arbitration awards, and PERB decisions.
- Translated complex economic and administrative datasets into executive-ready insights and decision memos, applying data visualization and scenario analysis techniques to support senior leadership negotiation and risk-management decisions.

Booz Allen Hamilton Inc **Mclean, VA**
Data Scientist *July 2023 - November 2024*

- Increased task force productivity by 10x and improved data accuracy by 50% by architecting a robust ETL system using SQL and Azure, enabling seamless integration of new product features and supporting scalable data science workflows.
- Enhanced decision-making for leadership and stakeholders by managing a team of six to develop interactive Tableau dashboards and KPIs, leveraging Python and Apache Spark for large-scale data processing and aligning technical solutions with business needs.
- Improved response time by 35% and expanded service capacity by 5,000 users monthly by redesigning and optimizing the Salesforce platform to monitor user engagement and gaming-like support features, utilizing data science methodologies to track LTV and other critical metrics.
- Boosted fraud detection accuracy and reliability for product reviews by developing and deploying NLP and machine learning models in a cloud-based analytics environment, using strong communication skills to present actionable insights to both technical and non-technical audiences.
- Facilitated adoption of modern BI tools and improved data accessibility by collaborating with stakeholders to pilot Looker dashboards and experiment with BigQuery SQL for performance optimization and future-proofing reporting infrastructure.

PROJECTS & OUTSIDE EXPERIENCE

Enterprise Cloud Analytics Capstone | University of Maryland *March 2024 - May 2024*

- Designed and implemented a cloud-native analytics platform using Apache Spark, Kafka, Databricks, and AWS/Azure to process large-scale batch and streaming enterprise data.
- Delivered risk-adjusted cost forecasting and sensitivity analysis, reducing decision uncertainty by ~30% and enabling faster, data-driven tradeoff evaluation across complex operational inputs.
- Built reproducible, scenario-based analytics workflows supporting multiple concurrent policy and cost scenarios, providing leadership-ready insights for enterprise decision support.

Amazon Fraudulent Review Detection using NLP & ML *May 2024 - August 2024*

- Designed and implemented an end-to-end fraud detection pipeline using Python, scikit-learn, and NLP, achieving 90.68% accuracy and 90.62 F1-score through TF-IDF–based feature engineering and robust model evaluation.
- Engineered interpretable behavioral and text-based features (e.g., sentiment polarity, extreme language usage, review length) to surface manipulative patterns, support model explainability, and inform risk and engagement analysis.
- Built scalable, analytics-ready datasets via production-style ETL and preprocessing workflows (tokenization, normalization, stemming, lemmatization), enabling downstream visualization, monitoring, and model performance analysis across BI tools and cloud analytics environments.

SKILLS

- **Programming & Core Analytics:** Python, SQL, R
- **Machine Learning & AI:** Machine Learning, Feature Engineering, Model Training & Evaluation, Time-Series Forecasting, Statistical Modeling, Natural Language Processing (NLP), Deep Learning, Scikit-learn, TensorFlow, PyTorch
- **Big Data, Cloud & Data Engineering:** Apache Spark, PySpark, Distributed Computing, Big Data Analytics, Streaming Data Processing (Kafka), Databricks, Cloud Computing (AWS, Azure, GCP), ETL / ELT Pipelines, Data Architecture & Schema Design
- **Databases & Data Warehousing:** Relational Databases, NoSQL (MongoDB), Graph Databases (Neo4j), Data Warehousing (Snowflake, Redshift)
- **Analytics, Visualization & Decision Science:** Financial & Business Analytics, Optimization & Operations Research, Data Visualization (Tableau, Power BI), Reproducible & Governed Analytics, Version Control (Git)